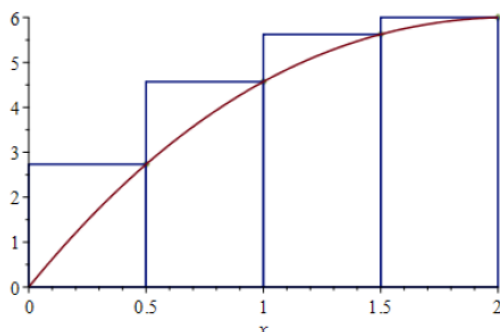


Sec 5.3 – The Fundamental Theorem and Interpretations

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In our first work to estimate distance from velocity, we found the area of rectangles to approximate the area under the curve:

time (s)	0	0.5	1	1.5	2
velocity (m/s)	0	2.7	4.6	5.6	6



$$\int_0^2 f(t) dt \approx f(0.5) \cdot \frac{1}{2} + f(1) \cdot \frac{1}{2} + f(1.5) \cdot \frac{1}{2} + f(2) \cdot \frac{1}{2}$$

What are the **units** of the integral? Why do you know?

Since $f(t)$ is in m/s and t is in s, the integral has units of $\text{m/s} \cdot \text{s} = \text{m}$. So this makes sense as a distance!

Partner up

Suppose our function $f(t)$ has one of the following meanings and units.

Then what are the units of the definite integral $\int_a^b f(t) dt$?

- ① $f(t)$ measures acceleration at time t seconds, measured in $\frac{\text{ft}}{\text{s}^2}$
- ② $f(t)$ measures the rate of change of the population of a colony of bacteria at time t hours in thousands of bacteria per hour
- ③ $f(t)$ measures the rate that water is leaking out of a bucket t minutes after noon in mL per minute

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Solution: The integral has units $\frac{\text{ft}}{\text{s}^2} \cdot \text{s} = \frac{\text{ft}}{\text{s}}$. (This is the units for velocity!)

- ② $f(t)$ measures the rate of change of the population of a colony of bacteria at time t hours in thousands of bacteria per hour

Solution: The integral has units (thousands of bacteria per hour) $\cdot$ (hours) = thousands of bacteria. (This is units for population!)

- ③ $f(t)$ measures the rate that water is leaking out of a bucket t minutes after noon in mL per minute

Solution: The integral has units (mL per minute) $\cdot$ (minutes) = mL. (This is units for the volume of the water!)

We noticed that if the integrand function $f(x)$ was the derivative of a function $F(x)$, then the units for the definite integral had the units of $F(x)$.

We also know that if we have a velocity function $f(t)$, then $\int_a^b f(t) dt$ gives us the *change in position* or *displacement* from $t = a$ to $t = b$. How can we write this in terms of the position function $F(t)$?

We can also write the change in position as $F(b) - F(a)$.

Theorem (Total Change)

$$F(b) - F(a) = \text{Total change in } F(t) \text{ from } t = a \text{ to } t = b = \int_a^b F'(t) dt$$

In words, the definite integral of a rate of change gives the total change.

Theorem (The Fundamental Theorem of Calculus (FTC))

If f is continuous on the interval $[a, b]$ and $f(t) = F'(t)$, then

$$\int_a^b f(t) dt = F(b) - F(a).$$

Partner up

Suppose $F(t) = t^3$. Then $f(t) = F'(t) = 3t^2$.

Evaluate $\int_1^2 3t^2 dt$ using the Fundamental Theorem of Calculus.

$$\int_1^2 3t^2 dt = F(2) - F(1) = 8 - 1 = 7$$

Note that to use the Fundamental Theorem to calculate the **exact** value of a definite integral, we need to know the antiderivative F . In the next chapter we will discuss techniques for computing antiderivative. And as we've done previously, we can **estimate** a definite integral with right or left-hand sums.

Partner up

Compute $\int_1^3 2x dx$ by two different methods.

Using left- and right-hand sums, we can approximate this integral as accurately as we want. With $n = 100$, for example, the left-sum is 7.96 and the right sum is 8.04. Using $n = 500$ we learn

$$7.992 < \int_1^3 2x dx < 8.008.$$

The Fundamental Theorem, on the other hand, allows us to compute the integral exactly. We take $f(x) = 2x$. We know that if $F(x) = x^2$, then $F'(x) = 2x$. So we use $f(x) = 2x$ and $F(x) = x^2$ and obtain

$$\int_1^3 2x dx = F(3) - F(1) = 3^2 - 1^2 = 8.$$

(5.3 #28) An old rowboat has sprung a leak. Water is flowing into the boat at a rate $r(t)$, given in the table.

time (min)	0	5	10	15
$r(t)$ (liters/min)	12	20	24	16

- Write an integral that gives the volume of water that has flowed into the boat during the 15 minutes.
- Compute upper and lower estimates for the definite integral.
- If there were already 25 liters of water in the boat at $t = 0$, about how many liters of water are there in the boat at $t = 15$ minutes?

a $\int_0^{15} r(t) dt$

- b In both cases, we use 3 intervals of 5 minutes each: 0-5 minutes, 5-10 minutes, and 10-15 minutes. The units of the definite integral are given by (liters/minute) · (minutes) = liters.

Upper estimate: Use the larger value from the table for each interval

$$r(5) \cdot 5 + r(10) \cdot 5 + r(10) \cdot 5 = 20 \cdot 5 + 24 \cdot 5 + 24 \cdot 5 = 340 \text{ liters}$$

Lower estimate: Use the smaller value from the table for each interval

$$r(0) \cdot 5 + r(5) \cdot 5 + r(15) \cdot 5 = 12 \cdot 5 + 20 \cdot 5 + 16 \cdot 5 = 240 \text{ liters}$$

- c The estimates for the definite integral are the *change in volume*, so these amounts are added to what is already in the boat. This means the amount of water in the boat is between $240 + 25 = 265$ liters and $340 + 25 = 365$ liters.

Time permitting:

Pollution is removed from a lake on day t at a rate of $f(t)$ kg/day.

Explain the meaning of the statement $f(12) = 500$.

On day 12, 500 kg/day of pollution are removed from the lake.

If $\int_5^{15} f(t) dt = 4000$, give the units of the 5, 15, and 4000, and then give the meaning of the definite integral.

Both 5 and 15 are days, and the 4000 is $\frac{\text{kg}}{\text{day}} \cdot \text{day} = \text{kg}$ of pollution removed, so this definite integral means 4000 kg of pollution is removed from the lake from day 5 to day 15.